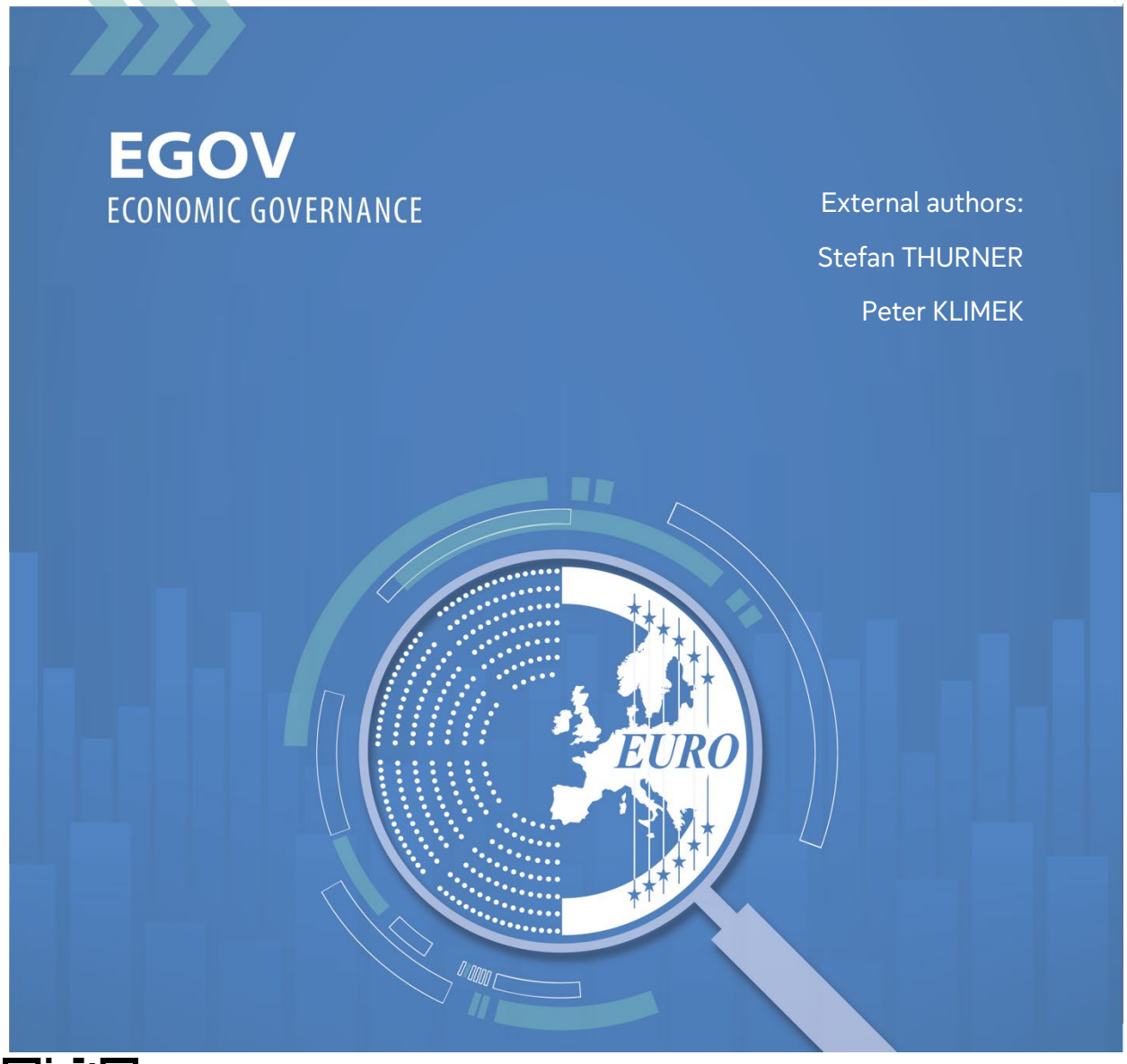
IN-DEPTH ANALYSIS

Requested by the ECON committee

European Parliament

Strategic dependencies, resilience and competitiveness in EU supply chains at the firm level

On the necessity to make EU supply chain information usable as a strategic asset



Economic Governance and EMU Scrutiny Unit (EGOV)
Directorate-General for Economy, Transformation and Industry
PE 779.855 – December 2025

EN

Strategic dependencies, resilience and competitiveness in EU supply chains at the firm level

On the necessity to make EU supply chain information usable as a strategic asset

Abstract

Technology now allows for complete, firm-level mapping of supply chains. The European Union can use this to build a Digital Twin of its economy, enabling it to simulate shocks, identify vulnerabilities, and enhance resilience. Developing this capability is essential for the EU's strategic autonomy as global competitors advance similar tools. We recommend establishing a dedicated legal and institutional framework to secure the Union's economic preparedness and competitive advantage.

This document was provided by the Economic Governance and EMU Scrutiny Unit at the request of the ECON Committee.

This document was requested by the European Parliament's Committee on Economic and Monetary Affairs.

AUTHORS
Stefan THURNER, Complexity Science Hub and Medical University of Vienna
Peter KLIMEK, Supply Chain Intelligence Institute Austria and Medical University of Vienna

ADMINISTRATOR RESPONSIBLE
Samuel DE LEMOS PEIXOTO
Harald STIEBER

EDITORIAL ASSISTANT
Donella BOLDI

LINGUISTIC VERSIONS
Original: EN

ABOUT THE EDITOR
The Economic Governance and EMU Scrutiny Unit provides in-house and external expertise to support EP committees and other parliamentary bodies in shaping legislation and exercising democratic scrutiny over EU internal policies.

To contact Economic Governance and EMU Scrutiny Unit or to subscribe to its newsletter please write to:
Economic Governance and EMU Scrutiny Unit
European Parliament
B-1047 Brussels
E-mail: egov@ep.europa.eu
Manuscript completed in December 2025
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LIST OF ABBREVIATIONS

CSDDD	Corporate Sustainability Due Diligence Directive
ESRI	Economic Systemic Risk Index
IO	Input-Output analyis
NACE	Nomenclature statistique des Activités économiques dans la Communauté Européenne is the industry standard classification system used in the EU
SME	Small- and medium-sized enterprise

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EXECUTIVE SUMMARY

Background – why understanding supply networks at the firm level matters

Europe's **efficiency, competitiveness, sustainability, resilience, and sovereignty** depend on the structure and dynamics of its **firm-level supply networks,** the dense webs of material, service, and knowledge flows connecting millions of EU firms. The dynamics of these networks show all dependencies, how efficiency and competitiveness emerge, how shocks spread, how innovation scales, and to what extent **sovereignty** is realized. Efficiency, competitiveness, sustainability, and resilience are **systemic properties**, not features of individual firms. Market forces optimize for short-term efficiency, not resilience or sovereignty. Therefore, **active, data-driven policy coordination** is essential to balance these objectives. Advances in **AI-based analytics, digital reporting, and administrative datasets**, especially **VAT transaction and customs records**, make it technically feasible to **reconstruct Europe's production networks in real time**. Understanding these networks allows the EU to:

- Identify **strategic dependencies** within Europe and beyond
- Make **competitiveness, risks and resilience** visible
- Reduce **systemic vulnerabilities**
- Conduct **simulation-based ex-ante policy testing**.

It is not only that traditional, sector-based economic data and input-output models cannot achieve these goals; worse, traditional models conceal the interdependencies that drive systemic risk and resilience and as a result are not fit for purpose to inform policy making.

Problem – The EU's strategic blind spot

Global powers, e.g. the **U.S., U.K., and China,** and major private institutions (e.g. **Bloomberg, J.P. Morgan**) are already building **national supply-chain intelligence systems** to monitor dependencies and risks in real time. The EU **lacks such capability** and therefore operates only with **partial awareness** of the structures that determine its economy. Without supply-chain intelligence, Europe risks **relying on foreign insights about its own economy**. In the data age, **knowledge of supply-networks is a form of geopolitical power**.

Solution – Establishing a European supply chain intelligence capability

To gain the necessary capabilities, the EU should **pool and harmonize national administrative data**, especially **VAT and customs records,** under a trusted institutional and legal framework.

Proposal: Establish a European Supply Chain Intelligence Institution: A secure, independent body to link data, develop analytical tools, and provide actionable intelligence for economic policy, crisis management, and sustainability. Its key functions:

- Build a harmonized European data layer from national resources

- Host centres for analytic excellence, data analytics, and simulation infrastructure
- Define safe, efficient, data-access and governance rules.

Positive side-effects: Reduce **VAT fraud** and fiscal leakage; Support **evidence-based, simulation-tested policymaking;** On a wider scale, also enable **low-bureaucracy supply chain due diligence regulations** and monitor **carbon flows and circularity** in real time.

Conclusion

Supply-chain intelligence is a **strategic necessity**. A **European Supply Chain Intelligence Institution** would convert **fragmented** data into **actionable foresight**, reinforcing Europe's **competitiveness, efficiency, resilience, sustainability, and sovereignty**. In the emerging data economy, **"who knows the networks, leads the economy".** Europe must act decisively now to turn knowledge into sovereignty.

1. SUPPLY CHAIN NETWORKS AT THE FIRM LEVEL: WHY THEY MATTER FOR THE EU

1.1. Supply chains as the metabolism of the European economy

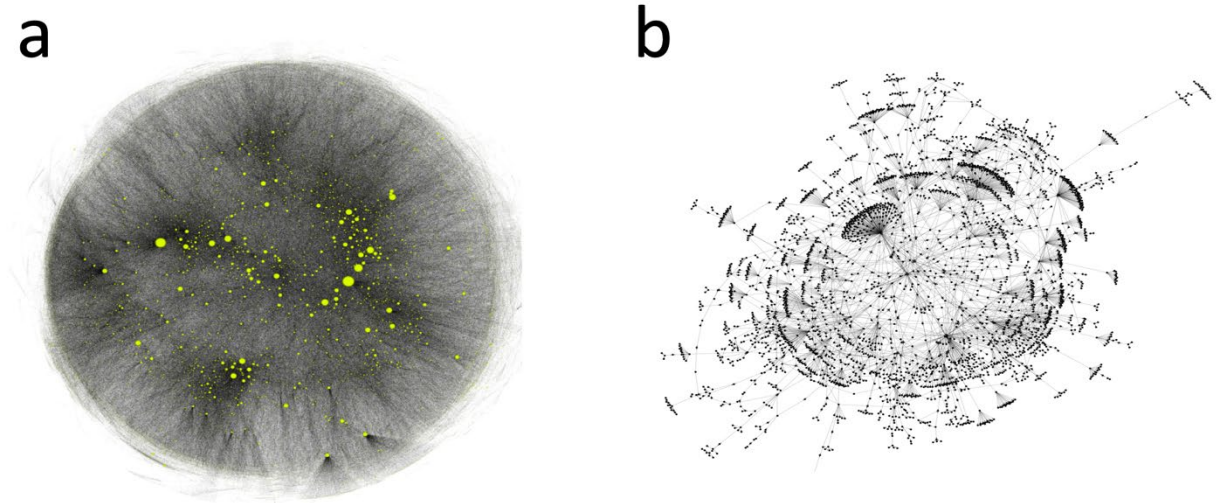
The European Union's most pressing economic vulnerabilities, ranging from industrial competitiveness to strategic autonomy, are rooted not in isolated sectors, but in the structure and exposure of its supply chains. Supply chains constitute the metabolism of modern economies: they embody the flow of materials, components, products, services and knowledge across millions of firms in the EU and underpin the functioning of every societal activity, from food supply to energy transition and defence production.

Yet despite this foundational role, our understanding of these supply chains remains remarkably limited [Pichler2023]. Policymakers still often rely on outdated models that depict supply relations as linear and controllable production sequences, step-by-step processes from raw materials to finished goods [Baldwin2023, Antras2021]. While such representations may still apply to individual production processes, they fundamentally mischaracterize how supply relationships operate at the system level.

1.2. From supply chains to supply networks

What appears as a linear value chain at the product level becomes, at the firm level, a dense and overlapping network, see Figure 1. Different products share suppliers, customers, logistics hubs and technologies, forming interlocking structures that generate not only synergies, efficiency, spillovers and redundancies—but also systemic vulnerabilities.

Figure 1: (a) Snapshot image of the supply chain network of the Hungarian economy as reconstructed from VAT data in 2019, showing about 100,000 firms (green dots, size is annual turnover) and 1.5million supply links. (b) Shows a blow-up with 500 randomly selected firms; Their individual supply links are visible; They are structured along supply chains; Supply chains all are connected to form a highly dynamic supply network.



Source: Images after [Diem2022].

Within a national economy, firms are connected through supplier–customer relationships, exchanging goods and services in real time. The number of firms captured in existing national supply network datasets can range from several hundred thousand in countries such as Belgium and Hungary to several millions in Japan, India and Brazil; see [Pichler2023]. Overlapping value chains form what we refer to as *supply networks*. Recent empirical research demonstrates that the structure of these supply networks is radically different from the simplified chains often assumed in policy debates [Bacilieri2023, Diem2022].

The average firm in Europe depends on 30–50 direct suppliers and customers [Bacilieri2023]. However, this dependency is extremely uneven. While small and medium-sized enterprises may operate within relatively narrow supplier circles, large firms can maintain supply relationships with tens of thousands of suppliers and way over one hundred thousand customers globally. A striking example is Volkswagen, which reports sourcing from more than 60,000 supplier locations spread across 100 countries[1], reflecting the complexity behind a single vehicle, which typically contains between 70,000 and 90,000 individual parts. Similar examples can be found for example in the production of semiconductors [Klimek2024].

1.3. Dynamic and fragile structure

Importantly, these networks are not static structures. They are constantly re-configuring. A recent large-scale firm-level study found that about one quarter of the firms and more than half of supply relationships dissolve within a single year, indicating that supply networks maintain a stable global "core" of long-term partnerships, surrounded by a highly dynamic "cloud" of transient relations [Reisch2025].

Recognizing supply chains as dynamic networks rather than fixed chains fundamentally changes how we understand efficiency, risks, competitiveness, innovation and resilience. While a network that can reconfigure rapidly is efficient, because it continuously removes redundancies, the same capacity for reconfiguration is also the basis of resilience, because it enables firms to rapidly re-establish supplies after disruptions. Efficiency and resilience are often seen as opposing goals, but they can be understood as different emergent properties of this network dynamic. This understanding is, in turn, the prerequisite for designing specific strategies and solutions that could potentially reconcile these initially opposing goals. One example would be strategies to incentivise circular business models that simultaneously reduce resource dependencies.

1.4. Strategic relevance for EU competitiveness and sovereignty

This is not merely of academic interest; it directly underpins current EU policy priorities. The EU Competitiveness Compass, informed by the Draghi report [Draghi2024] on the future of European

[1] https://www.volkswagen-group.com/en/press-releases/volkswagen-group-fourth-responsible-raw-materials-report-published-18393

competitiveness, identifies three key imperatives. Each of them ultimately requires significant rewiring and upgrading Europe's supply networks:

- **Closing the innovation gap**: Creating a fertile environment for AI, quantum, biotech, advanced materials and space technologies requires not only supporting startups, but building *entirely new supply networks* that enable scaling, integration and industrial deployment across sectors.
- **Decarbonizing the economy**: The shift to low-carbon technologies and circular business models (such as steel scrap recycling or low-carbon cement production) demands restructuring of supplier relations, access to new input materials, and coordination across all industrial tiers.
- **Reducing dependencies and diversifying suppliers**: Strategic autonomy cannot be achieved by stockpiling raw materials alone It requires the diversification of supplier networks and the creation of technological innovations through new supply networks that reduce dependencies.

Europe's efficiency, competitiveness, resilience and capacity to innovate all emerge from the structure and dynamics of its supply networks. They are not fixed infrastructures but constantly adapting. The ability to enable, guide, monitor and accelerate and not hinder that adaptation will determine whether Europe remains a global standard-setter—or becomes structurally dependent on regions who do, see also section 5.

1.5. Why supply network governance is a strategic priority for the EU

Efficiency, competitiveness, resilience and sustainability are often discussed as separate policy goals, yet they are emergent dynamic and structural properties of the same supply network. Consequently, policies that aim to optimise the network for one of these goals will generally not lead to improvements in the others, unless they are carefully designed to do so. In other words, these concepts are *not* properties of individual firms or sectors but rather *systemic features* that emerge from how supply networks are structured and how they evolve over time. A supply network optimized purely for efficiency, e.g. by concentrating production in a few low-cost hubs, will tend to sacrifice resilience, as redundancies are eliminated, and flexibility is lost. Conversely, a system built for high levels of resilience, e.g. through localized buffers, diversification or strategic redundancies, will typically incur higher short-term costs and appear "inefficient" by conventional accounting. Hence, there is a trade-off between efficiency and resilience, unless supply chains are carefully engineered [Choi2023, Lücker2025].

The conclusion is unavoidable: resilience and sustainability will in general *not* emerge automatically from market forces. The situation is comparable to the management of systemic risks in financial markets. In the wake of the 2007/8 financial crisis, macroprudential policy tools were developed to improve the system's overall resilience [Hanson2011, Leduc2017]. We argue that, in supply networks, firms optimize profit and minimize costs and revenue, but not for resilience, strategic autonomy, or social stability. Left to themselves, firms have little incentive to pay for redundancies from which others will benefit. Nor can they, on their own, correct structural vulnerabilities that are embedded firm-level network structures.

This is because resilience is not an attribute of individual firms but of the entire network. A company's exposure to disruption depends not only on its *own* behaviour but on the behaviour of its *suppliers* and even on the relationships among *its suppliers' suppliers.* Much like in epidemiology, where individual risk of getting infected depends on the connectivity of others, the resilience of firm A may be determined by how firms B and C transact with each other, relations which A neither sees nor controls.

The transition to a *circular economy* offers another clear illustration of the necessity of a systemic network-based view. Reuse and recycling of materials, central to EU strategies for decarbonizing steel, reducing critical raw material dependencies and enabling green growth, cannot be achieved by single firms acting in isolation. Every company along the chain must *jointly* accommodate end-of-life recovery. Without coordination at the systems level, circularity simply will neither emerge, nor scale.

Consequently, neither individual corporate action nor market mechanisms alone can deliver resilient and sustainable supply networks. This is simply because network information is not available to firms and markets and thus quantitative levels of systemic risk and resilience are not obtainable; the situation is similar as in financial networks [Poledna2016]. Active policy design is required, not to micromanage firms, but to embed new incentives, shape economic coordination and make structural and strategic risks visible.

However, policy is currently operating in the dark. Even as the EU develops an ambitious regulatory agenda—ranging from the Corporate Sustainability Due Diligence Directive (CSDDD) and the Critical Raw Materials Act, to CBAM, the Net Zero Industry Act, the EU Battery Regulation, the Forced Labor Regulation and IMERA—systemic risks (outside the financial sector) remain under-appreciated due to the limited understanding of supply networks. The recent experience with CSDDD serves as a cautionary tale. Initially celebrated as a transformative step toward responsible supply governance, the directive soon faced backlash for being disconnected from the true complexity and dynamism of real supply networks. As a result, its scope was reduced, thresholds raised and implementation postponed, creating confusion while weakening the original ambition.

The lesson is clear: the EU cannot afford to design supply chain policy without supply chain intelligence. If we want to manage efficiency, resilience and sustainability in a coordinated manner and at the same time ensure competitiveness, we must first *see* the system we are trying to govern. Only then can interventions be calibrated and monitored to achieve their intended effects, rather than producing administrative burden with limited impact.

To boost EU competitiveness while safeguarding strategic autonomy, we must learn to observe, simulate and steer supply networks as systems on the firm-level. This is no longer a purely economic or technical challenge—it is a necessary *foundational governance capability for ensuring European sovereignty* in the data age.

2. MAKING SUPPLY NETWORKS VISIBLE

2.1. The supply chain knowledge paradox

Europe finds itself in a paradoxical moment when it comes to understanding its supply networks. On one hand, our knowledge has increased more rapidly in the past decade than in the previous hundred years, driven by the emergence of unprecedented datasets, digital tools and conceptual progress to derive practical information from them. On the other hand, this newfound visibility is not available and usable to those that would most need it. Even most companies lack clarity on their own supply chains. Only 6% of firms report having supply chain visibility—knowledge of physical, logical, or financial flows—from suppliers of their suppliers, through production, to customers of their customers. In contrast, close to 80% have restricted visibility, covering no more than two of these steps[2]. Even global leaders often remain unaware of who supplies their suppliers—or how disruptions in distant upstream tiers may cascade into their own operations. As a result of this poor visibility, not only policy makes but also industry stakeholders operate with partial awareness of systems that determine strategic autonomy, industrial resilience and crisis readiness.

2.2. From sectoral aggregates to firm-level networks

A century ago, the highest level of economic visibility available to policymakers (via official statistics?) was the sectoral classification of national economies. The development of input–output tables in the 20th century allowed economists to approximate how different sectors interact with one another. Over time, these tables became more granular, evolving into multi-regional input–output models by the 1990s, disaggregating global production into a few thousand sectoral categories.

This was a major leap forward—but still the granularity of the data remained coarse. With the rise of big data, a new revolution has begun, and firm-level supply network mapping [Borsos2020, Bacilieri2023, Diem2022] is possible. In the past decade, datasets have emerged that track millions of companies and their supplier–customer relationships [Pichler2023]. This evolution is about to transform economic analysis from abstract aggregates to real-world industrial cartography on the firm-level, in principle enabling policymakers to trace dependencies and vulnerabilities not just across sectors, but *across specific corporate dependencies*. It has been shown how much predictive power can be gained by shifting from a traditional input–output approach to granular firm-level data [Diem2024]; see also section 3.2.

2.3. Data sources driving the supply network revolution

There are three main categories of datasets powering this visibility revolution: commercial intelligence datasets, administrative datasets and AI-augmented information extraction systems. See Annex for details.

[2] GEODIS, 2017 Supply Chain Worldwide Survey, https://geodis.com/sites/default/files/2019-03/170509_GEODIS_WHITE-PAPER.PDF

2.4. The emerging role of artificial intelligence

A new frontier in supply network mapping is now unfolding. AI systems can increasingly replicate (and massively scale) the work traditionally performed by commercial supply chain analysts. Instead of relying on structured filings, AI agents can extract supplier–customer relationships from unstructured text, such as press releases, news articles, regulatory disclosures or corporate websites [Almahri2025].

These AI-augmented methods do not replace administrative or commercial data; rather, they enrich them, adding semantic detail about the nature of each relation: whether it concerns raw materials, logistics, contract manufacturing, digital services, or technology licensing. When combined appropriately (considering the known problems of validation, hallucination, accuracy and energy and infrastructural requirements) with administrative transaction records, this hybrid approach offers for the first time a realistic path toward full-spectrum monitoring of complex industrial systems.

3. NEW POSSIBILITIES: DATA-DRIVEN TOOLS FOR AWARENESS, PREPAREDNESS AND PLANNING

Until recently, supply chains were primarily observed through aggregated national input-output tables or sectoral trade statistics. These approaches, while being one of the working horses in economics, mask most of the relevant complexity, heterogeneity and dynamics of the value chain. Today, with the increasing availability of firm-level transaction data (e.g. derived from VAT records or electronic invoicing), it has become possible to reconstruct supply networks at national scales and analyse and monitor them on the firm-level. However, mapping who trades with whom is only a first step. To inform policy, one must also understand *how* the entire *networks* function, how they respond, amplify or absorb shocks, and how interventions, disruptions and innovation affect and reshape them.

Because supply networks on the firm-level are way too large, dense and dynamic to interpret by intuition or static visualization alone, *network science* and *simulation-based policy testing* are emerging as indispensable tools for economic governance. The following advances illustrate how efficiency, competitiveness, systemic risk and resilience can now be made *quantifiable,* turning them into *measurable system* properties that can be used for policy.

3.1. Making dependencies visible

Competitiveness is not merely a function of individual firms' innovation capacity and productivity but of *who depends on whom*. Dependencies in supply chains translate into dependencies in economic power, trade leverage and vulnerability to foreign disruptions. A striking illustration of this comes from the Great East Japan Earthquake of 2011, which triggered a tsunami and the Fukushima Daiichi nuclear disaster. Although the shock was geographically localized, the *economic consequences were systemic and far-reaching*, offering a so-called natural experiment for studying how risk is transmitted along supply networks [Carvalho2016].

Research showed that negative shocks propagate in two directions. *Downstream*, disrupted firms reduce deliveries to their customers, constraining not only direct buyers but also *their* customers further down the chain. *Upstream*, disrupted firms also reduce demand from their suppliers. However, upstream propagation can lead to either further losses or temporary gains for initially unaffected companies, as suppliers of disrupted firms may reallocate spare capacity elsewhere. In both directions, the shock intensity declines with distance but remains significant across multiple tiers.

The quantitative effects were substantial. One study estimated that the shock propagation through supply linkages accounted for a 1.2 percentage point decline in Japan's gross output in the year following the earthquake, even among firms with *no direct connection* to the disaster zone. Another study, using a different model [Inoue2019], found that *indirect* effects on firms not directly damaged amounted to 2.3% of GDP, *more than 100 times larger than the direct losses*.

Crucially, network structure amplified the shock. A small number of firms had orders of magnitude more supply links than the average company. Simulation analysis confirmed that without this extreme heterogeneity, indirect amplification would have been *only one-tenth* of its actual size. Details matter. Strategies to increase competitiveness and resilience must be based on detailed *real-world supply network structures* in real time, not on aggregated sector averages or stylized models.

3.2. Making systemic risk and resilience visible and quantifiable

Systemic Risk. While studies on shock propagation reveal *how disruptions spread*, another recent line of work aims to measure *which firms represent critical systemic risk*. [Diem2022] developed a methodology to compute an *economic systemic risk index* (*ESRI*) for every firm in a country, using supply transaction data from VAT records combined with firm-level production functions.

The approach is conceptually simple but computationally powerful. For each firm, its temporary failure is assumed and simulated, meaning it supplies nothing to its customers and demands nothing from its suppliers. It is then observed how the virtual default of that firm propagates the resulting loss through the entire network and keeps track of how losses accumulate. At each step, upstream and downstream firms adjust their production levels based on lost inputs or lost demand. The *ESRI* of a firm is defined as the *fraction of total national output likely to be affected* by its failure.

When applied to nation-wide firm-level supply networks, the results were both alarming and instructive. In Hungary, only 32 companies formed a *systemic cluster ("systemic risk core")*—each with an *ESRI* around 23% of national output [Diem2022]. If *any one* of these firms failed without immediate substitution, nearly a quarter of the economy would be immediately affected. Further, 63 firms had *ESRI* above 5%, and 165 firms above 1%. These firms spanned sectors such as energy, electrical equipment, chemicals, electronics, vehicle manufacturing and machinery repair. Similar systemic risk cores have been reported also for specific supply networks in other countries [Zelbi2025]. It is reasonable to assume that the existence of a "systemic risk core" holds generally also for larger EU economies. It is one of the policy recommendations to become able on an institutional level to make these systemic risk cores visible and monitor them, see section 5.3.

Perhaps most surprisingly, firm size was *not* a reliable predictor of systemic importance [Diem2022]. While many critical firms were large, some were relatively small enterprises, however, embedded in structurally central positions. What distinguished highly systemically relevant firms was not revenue, but their *position* in the network. Roughly two thirds of the most systemically relevant firms formed a strongly connected component: a tightly interdependent cluster where the failure of one would imply that all those in the cluster would no-longer be able to sustain their production. In such a situation, systemic damage can only be avoided by immediate substitution of the failing company from outside the national economy, which creates systemic dependencies on other economies in times of disaster.

This demonstrates a profound insight: systemic importance is not the same as economic size. A small firm embedded in a bottleneck position may be more critical than a large but redundant one. Generally, the number of firms in such bottleneck positions will increase when efficiency increases and the number of competitors shrinks. To what extent that is true for individual firms can only be addressed when seeing the position of the firm within the wider network. Without visibility into these structures systemically important firms cannot be identified correctly and thus policy effectively cannot prioritize, protect or regulate systemic actors. While for large companies it might sometimes be possible to identify bottleneck companies within their supply chain and try to internalize supply risks along the chain [Mukunde2022], this is in general not possible for individual firms.

Resilience. Resilience in supply chains cannot be meaningfully assessed without considering *who* is connected to *whom*. [Diem2024] demonstrated that resilience becomes visible and quantifiable *only* when the actual firm-level network structure is explicitly considered. Aggregating firms into sectors, such as in traditional Input-Output (IO) analysis, erases the heterogeneity of real supply relationships, loses critical details and leads to fundamentally misleading conclusions [Diem2024].

This study showed that firms grouped under the same industry classification are extremely diverse in terms of *where* they buy and *where* they sell. On average, two firms classified within the same industry purchase only 23.5% of their inputs from the same industries and sell only 19.3% of their outputs to the same sectors. In other words, firms within one sector are *not* interchangeable in terms of their actual supply dependencies. As a result, sector aggregation leads to misestimations of shock propagation of up to 37.1% at the aggregate level, and between 59.5% and 150% for individual industries. This partially also explains the known limitations of current Input-Output analyses in economic forecasts and *ex ante* policy evaluation.

This finding has profound implications. The *magnitude* of economic losses from disruptions such as COVID-19, the war in Ukraine or natural disasters depends *crucially on which exact firms* absorb the initial shock. Even two firms in the same region or industry can respond entirely differently, and their failure can cascade through completely different parts of the network. Traditional industry-level tools are *structurally incapable* of representing such tail risks, especially when multiple *systemically important firms* are jointly affected [Acemoglu2015].

This limitation extends far beyond shock propagation. Any policy analysis involving growth, inflation dynamics, emissions accounting or price pass-through will be fundamentally distorted when conducted

at the sector level. For example, in the context of climate action, attempts to estimate *scope 3 emissions* (indirect emissions embedded across supply chains) produce major errors when based on aggregated Input-Output data. Firms within the same NACE classification use *different inputs*, sell to *different destinations* and therefore carry *vastly different emissions footprints*. Accurate and *targeted* decarbonisation strategies will thus *require firm-level data*, not sector averages; see e.g. [Stangl2025].

For a detailed showcase for how positive side effects of supply chain intelligence can be derived from in silico policy testing, see Annex 2.

4. STRATEGIC IMPERATIVE: BUILD NEW TECHNICAL CAPABILITIES OR RISK FALLING BEHIND

The ability to map, monitor and analyze supply chain networks at scale is no longer a theoretical possibility, it is rapidly becoming a *strategic capability*. The question for the European Union is no longer *whether* supply chain knowledge will shape competitiveness, security and industrial capacity, but *who* will possess that knowledge, and for whose benefit it will be used.

While research institutions and statistical agencies in Europe are only beginning to unlock the potential of administrative data sources such as VAT or transaction data, private actors and foreign governments are moving ahead at high speed. Large financial institutions such as *Bloomberg*[3] and *JP Morgan*[4] are already investing heavily in reconstructing global supply chain networks using proprietary data, alternative data scraping, AI-based extraction from financial reports and real-time trade flows. Their objective is clear: to gain exclusive foresight into bottlenecks, supplier dependencies and corporate vulnerabilities—and to monetize this information through financial products, risk pricing and strategic advisory services. In this scenario, Europe risks becoming a mere *customer* of strategic intelligence about *its own supply chains*.

At the same time, foreign governments are taking decisive steps to build public infrastructures for supply chain intelligence. The United Kingdom's *Global Supply Chain Intelligence Pilot*[5] exemplifies this shift. Its goal is to integrate customs data, company registries, geospatial information and transaction records into a *centralized system for real-time risk monitoring*, enabling the government to identify critical vulnerabilities before crises hit. Early reports suggest that the UK aims to operationalize this as a *permanent institution*, linking industrial strategy, trade defence, national security and emergency response.

[3] Bloomberg offers supply chain intelligence with the SPLC function in the Bloomberg terminal, https://www.bloomberg.com/professional/solutions/corporations/supply-chain/

[4] JP Morgan recently announced strategic research efforts in the company level in supply chains with a focus on strategic areas like rare earths, AI and technology within its recent decision for a 1.5 trillion USD investment in a strategic resiliency initiative https://www.jpmorganchase.com/newsroom/press-releases/2025/jpmc-security-resiliency-initiative?utm_source=newsletter&utm_medium=email&utm_campaign=newsletter_axiosam&stream=top

[5] UK's Global Supply Chain Intelligence Pilot https://www.gov.uk/government/publications/global-supply-chain-intelligence-programme-gscip-privacy-notice/global-supply-chain-intelligence-programme-gscip-privacy-notice

The United States is pursuing similar ambitions under the frameworks of the *Defense Production Act*[6], the *Industrial Base Analysis and Sustainment Program*[7], and initiatives such as the *Supply Chain Disruptions Task Force*[8]. These programs increasingly rely on combined intelligence from federal agencies and private data aggregators to monitor dependencies in semiconductors, pharmaceuticals, batteries, minerals and critical infrastructure.

Meanwhile, China has been systematically gathering firm-level supply chain data for over two decades, supported by state-linked credit rating agencies and digitized tax and customs regimes. This data underpins not only economic planning but also export controls, dual-use industrial policies and strategic dependency leverage. In 2018, China launched the supply chain innovation and application pilot policy to promote supply chain digitalization[9]. In practice, China might know far more about the structure of European supply chains than Europe does about its own, as EU companies need to apply for licenses to import critical raw materials from China, thereby disclosing their material uses and production processes.

The consequence is clear: digital supply chain intelligence is emerging as a core instrument of geopolitical power, much like energy security was in the 20th century. Control over data on *who depends on whom, for what, and with what alternatives* defines strategic leverage. Failing to develop the capacity on the EU level to *see* and *simulate* these dependencies means that Europe will be forced into a permanently reactive stance, responding to crises after they materialize and after others have reacted, rather than shaping outcomes in the EU's interests in advance.

In short: Supply chain visibility is no longer merely an analytical question. It is a question of sovereignty. Knowing the own supply chains—better, faster and more accurately than others—will determine whether the EU *leads the global transition*, *follows under pressure,* or *falls into strategic dependency*.

Europe cannot afford to lag behind. The tools now exist, in the form of administrative datasets, AI extraction methods and network-based policy modelling, to build a *European Supply Chain Intelligence System* capable of supporting regulation, industrial strategy, crisis prevention and decarbonization at the same time. But these tools must be institutionalized, not left to fragmented pilot projects or private

6 Defense Production Act https://www.fema.gov/disaster/defense-production-act

7 Industrial Base Analysis and Sustainment Program https://www.businessdefense.gov/ibr/mceip/icam/index.html

8 Supply Chain Disruptions Task Force https://www.congress.gov/crs_external_products/IN/PDF/IN11927/IN11927.1.pdf#:~:text=Supply%20Chain%20Task%20Force%20On%20June%208%2C,directed%20to%20increasing%20data%20sharing%20among%20agencies.

9 The policy aimed to promote the digitalization, intelligentization, and green development of supply chains, which research has used as a quasi-natural experiment to study its effects on corporate performance. [Qing2025] found the policy significantly improves corporate competitiveness, green innovation, energy performance, and sustainability.

monopolies. The EU can build on a body of existing supply chain expertise in institutions. Such as e.g. ASCII[10] in Austria and central banks, e.g. in Hungary [Borsos2020].

4.1. Necessity of collective European action

If supply chain intelligence is to become a strategic capability for Europe, it cannot remain fragmented at the national level. A coordinated European effort is not just desirable, it is structurally necessary. No member state alone, regardless of size or economic power, can independently see or manage the full web of dependencies that determine Europe's economy. The strength of the single market lies in its interdependence — and so does its vulnerability.

National maps are emerging—but they stop at the border

Several member states already have the possibility to reconstruct temporally resolved national supply chain networks from VAT transaction data, customs declarations, industry registries or financial reporting. Some have already operationalized firm-level production network models for systemic risk assessment [Diem2022].

However, no national network is sufficient on its own. Modern manufacturing networks do not align with borders. A "national view" of supply dependencies is inherently incomplete and misleading, because critical links (especially in intermediate goods and logistics) often lie across the border. A firm considered "secure" in a national dataset may turn out to be highly vulnerable when its foreign second-tier suppliers or distribution partners are revealed. Without a cross-border perspective at least on the EU level, risk assessments will typically underestimate dependencies and misidentify priorities.

Why merging national data is essential—and why it is currently impossible

To obtain a true European supply chain intelligence capability, national networks must be merged into a single, interoperable data structure. Technically, this is now feasible: EU-wide harmonized identifiers such as VAT numbers and customs registration codes exist and allow cross-country linkage. Methodological standards to align classifications and transaction thresholds are already available from trade statistics and Input-Output accounting.

However, the current legal status quo *prohibits* Eurostat and national statistical offices in the member states from doing so, see for instance Regulation (EC) No 223/2009 on European Statistics which establishes the concept of statistical confidentiality. Such existing data-sharing regulations were designed for aggregated economic indicators, not for strategic monitoring of inter-firm dependencies. These regulations forbid the exchange of granular firm-level relational data, even for clearly defined public interest objectives such as resilience, crisis planning or industrial strategy. Without explicit *European legislation* enabling and establishing institutionalized data pooling and harmonization between the member states for purposes such as surveillance by public authorities, Europe will fall

[10] The Supply Chain Intelligence Institute Austria (ASCII), enabled by the Federal Ministry of Labor and Economy in 2023, is Austria's centre for supply network intelligence, https://ascii.ac.at/

behind private finance actors and foreign governments who do not face such institutional fragmentation.

Why high-quality, standardised data is critical

Supply chain intelligence is only as good as the precision of its underlying data. Incomplete, outdated or poorly classified linkages lead to false conclusions, either overestimating risks (triggering unnecessary interventions) or missing critical nodes (leading to blind spots of true risks). For this reason, harmonized high-quality and secure input data is a prerequisite.

There is a need for a European standard for supply chain data quality, covering entity resolution, transaction reconciliation and linkage verification, to ensure comparability across member states and sectors. Without it, merged datasets risk becoming inconsistent mosaics rather than coherent infrastructures.

Ensuring safety, trust and compliance with GDPR

Finally, **privacy and European data protection standards must be used as design principles**. Supply chain monitoring can be conducted without exposing sensitive firm identities to unauthorized actors. Techniques such as federated computation, pseudonymization, secure multi-party simulation and controlled access protocols are already in use, e.g., in the health and finance sectors, in some cases for decades. There is no reason why supply chain intelligence should not meet the same standards.

A **European Supply Chain Intelligence System can—and must—be built in a GDPR-compliant, secure manner**, where:

- Companies are **strictly protected against misuse** of their data
- Governments, selected EU institutions and defence gain **legitimate analytical capacity** for risk management
- Research gains **strictly controlled access** for public interest innovation.

Europe has the institutional experience to build such trusted infrastructures, as demonstrated by the European System of Central Banks or the EU Vaccines Joint Procurement Mechanism. The question is simply whether the EU acts soon enough.

5. POLICY RECOMMENDATIONS — BUILDING A EUROPEAN SUPPLY-CHAIN INTELLIGENCE CAPABILITY

Europe must move from *ad-hoc* pilots and fragmented national projects to a permanently operating, EU-level capability that can monitor, interpret and responsibly share firm-level supply-chain information for policy making. The recommendations below describe the institutional mandate, the data plumbing, and the safeguards required to turn a technical possibility into strategic public value. They are designed to be operational: each step both creates immediate utility and lays the groundwork for progressively richer and safer services.

5.1. Establish a trusted EU supply-chain intelligence institution

We recommend establishing a dedicated, independent EU institution charged with assembling, operating and interpreting a European Supply-Chain Intelligence System. Member states would report pseudonymized VAT payment records (firm A → firm B) to this body under strict legal safeguards, or alternatively grant it controlled access to the EU-wide electronic invoice system at the firm-level. The institution should report to a democratic body such as the European Parliament and be accountable to a board representing member states, the Commission, and a body of independent experts. Its legal mandate must be explicit: to collect, link, quality-control and analyse inter-firm transaction data for public European interest. It makes this information available to the respective institutions and government bodies. This will include industrial strategy, crisis prevention, regulatory compliance and research. This mandate must be underpinned by a clear legislative act that authorizes data flows across jurisdictions so that the institution can operate with legal certainty and authority.

5.2. Pool administrative datasets into a harmonized European data-layer

The new institution should operate a high security data platform that links national VAT transaction data, electronic invoices with customs records, intra-EU VAT exchanges and other administrative sources (business registers, customs, trade in goods data) at high temporal resolution. Where daily-resolution inter-EU VAT or customs feeds already exist, they should be ingested to give near real-time situational awareness. One objective is harmonization: resolving entity identities across countries, reconciling currencies and tax periods and standardizing product and service classifications so that analyses are comparable across member states. The new institution should be designed to operate in a federated mode at first, pulling metadata and running distributed queries, with a progressive pathway to more centralized analytics where appropriate and legally authorized.

5.3. Establish and host centres of analytic expertise

Data without analytic capacity is a limited resource for policy. The institution must recruit and convene multidisciplinary teams, expert in data- and network science, industrial economics, trade statistics, data engineering and secure large-scale computation. It should form long-term partnerships with existing research centres and initiatives (for example centres that already study production networks, economic

systemic risk, or supply-chain resilience) and foster competitive public research grants to innovate methods for mapping, simulating and scoring supply-chain vulnerabilities. The institution should also develop operational toolkits for ministries and regulators, including dashboarding, scenario simulation and impact evaluation modules that allow *ex-ante* testing of policies.

5.4. Gatekeeping, access control and accountability

Because most datasets are highly sensitive, the institution must operate a robust governance framework for data access and use. This framework must specify which institutions have access which outputs, for what purposes and under which legal and ethical constraints. Relevant actors (EU institutions, member state ministries, defence authorities, statistical offices and approved research teams) should be granted tiered, auditable access, aligned with clear purpose limitations and under a minimal-data principle. Independent oversight, including parliamentary reporting, a data-protection body, and a broad ethics advisory board are integral to the new institution's charter. All access should be logged and subject to regular audits.

5.5. Produce anonymized and aggregated public goods

The institution should routinely publish anonymized, aggregated indicators and risk maps that are useful for EU companies, regulators and the public while strictly protecting firm confidentiality. Examples of public outputs include regional systemic risk heatmaps, sectoral dependency indices, anonymized supplier concentration metrics, and simulated shock scenarios at aggregated spatial and sectoral levels. Such outputs democratize insights, reduce information asymmetries, and enable SMEs and policymakers to act without exposing individual firms' sensitive transaction histories.

5.6. Secondary benefits

VAT-fraud detection and fiscal integrity

A carefully governed data platform will also substantially improve public finance oversight. Linking payment and invoice streams across borders can significantly strengthen detection of VAT evasion and carousel fraud11, improving revenue collection and reducing distortions that harm compliant businesses. This fiscal benefit can be articulated as part of the institution's value proposition and may help finance initial deployment.

Support and simplify corporate sustainability due diligence

Firm-level transaction data is precisely the information companies struggle to obtain when complying with supply-chain due-diligence obligations. By enabling supplier certification, automated cross-checks against negative lists, and the generation of verified supplier profiles, the proposed EU institution can make CSDDD compliance way more effective and practically free of burden for millions

[11] [Braml2022] estimates that a significant fraction the EU's self-surplus puzzle, amounting to 307 billion EUR in 2018 (about 1.9% of the EU's GDP), is due to fraud in VAT.

of EU companies and businesses. Where automated checks flag potential human rights violations and other risks, companies can be directed toward targeted verification rather than forced into blanket, costly audits. This reduces administrative overhead and improves the real enforcement of human-rights and environmental standards.

Track carbon flows and enable targeted decarbonization policy

The same infrastructure enables robust and comprehensive tracking of embedded carbon flows through supply chains. By linking transaction volumes with firm-level emissions inventories (scope-1 and 2) and supply flows, the institution can generate credible scope-3 estimates and identify leverage points where emissions reductions produce the largest system-wide benefits. This capability supports supply-chain-sensitive decarbonisation instruments, such as differentiated CO_2 levies, targeted subsidies for low-carbon input alternatives, or tailored industrial transition plans, that can cut emissions at minimal social cost.[12]

5.7. Implementation and sequencing

A phased approach in the implementation of a harmonized EU-wide supply network infrastructure reduces legal and operational risk. An initial phase of not more than one year should focus on a small number of member states volunteering VAT data harmonization under pilot arrangements, building the technical stack and governance models and delivering demonstrable public goods such as aggregated risk dashboards and VAT fraud analytics. A second phase then expands country coverage and data inputs, introduces federated query capabilities for ministries and regulators and possibly pilots automated CSDDD checks and carbon-flow analytics. A final phase moves to broader operational integration—subject to full legal authorization—including real-time feeds and secure scenario simulation for crisis response.

Funding for the initial implementation could include EU grants (e.g., Horizon/REACT-style instruments), member state contributions and re-invested fiscal gains from improved tax compliance. Equally important are the necessary legal instruments: the European Parliament and Council must provide a clear legal basis for the new institution and its controlled sharing and protected use of critical data.

5.8. A strategic choice for Europe

Creating a trusted *European Supply-Chain Intelligence Facility*[13] is not merely a technical add-on to existing policy. It i an essential infrastructure for sovereignty in a world where supply-chain knowledge confers economic and geopolitical advantage. When done correctly, the institution will pay for itself through improved crisis preparedness, smarter regulation, better decarbonization outcomes and stronger fiscal integrity, while protecting firm confidentiality and civil liberties. Europe must decide

[12] Such capabilities could also serve other policy areas such as climate and employment policies in addition to macro-fiscal policies. Annex 2 provides further detail on such positive side effects of those data modelling capabilities.

[13] Such *Facility* could either be a new agency or be run by existing EU bodies such as Eurostat.

whether to build this capability on public, accountable terms or allow this strategic function to be concentrated in private actors and foreign states. The choice has long-term consequences for competitiveness, resilience, democratic governance, defence and sovereignty.

CONCLUSION—BUILDING STRATEGIC FORESIGHT FOR EUROPE'S SUPPLY NETWORKS

The global economy is entering an era where visibility into supply networks on the firm-level determines resilience, competitiveness and strategic autonomy. As recent shocks, ranging from the COVID-19 pandemic to the war in Ukraine and the semiconductor shortage, have shown, the ability to anticipate and mitigate disruptions depends not only on knowing *what* economies produce, but *how* that production is interconnected. Yet, despite its economic weight and technological sophistication, Europe remains largely blind to its own supply-chain structures and thus its risks and international dependencies.

This report emphasized that the tools to change this situation do exist today. Firm-level transaction data, whether from VAT records, electronic invoicing systems, or payment networks, can be transformed into dynamic maps of national and EU-wide supply networks near real-time. Combined with new analytical methods from data- and network science as well as artificial intelligence, such data enable a profound leap forward in understanding economic interdependencies, systemic risk and resilience. The scientific literature reviewed here has demonstrated that key economic properties—efficiency, competitiveness and resilience—emerge not from isolated firms or sectors but from the *networked architecture* of production itself. Policies that ignore this structure will underestimate both vulnerabilities and opportunities by orders of magnitude.

Europe is at a crossroads. Other regions are rapidly developing supply-chain intelligence infrastructures. The United Kingdom's *Global Supply Chain Intelligence Pilot*, U.S. efforts under the *CHIPS and Science Act* and the *Supply Chain Resilience Office* and China's growing investment in national industrial graph databases all point to a strategic recognition: in the twenty-first century, *supply-chain data are national security data*. If Europe fails to build its own capabilities to put them into use, it risks strategic dependence not only on foreign suppliers but also on foreign knowledge about its own economic systems.

The way forward requires collective European action. National initiatives using VAT or administrative data have proven technically and legally feasible, but their true value emerges only when connected into a coherent, cross-border view. The proposed *European Supply-Chain Intelligence Facility* would serve this purpose: pooling data, ensuring legal compliance and cybersecurity, and building, growing and providing analytical capacity for policymaking, crisis management and research. This institution would transform fragmented national efforts into a coherent European strategic asset.

To summarize, the benefits are far-reaching:

- **For economic policy**, it allows real-time monitoring of all industrial dependencies and the rapid evaluation of interventions, within and beyond Europe
- **For crisis management**, it offers early-warning capabilities and evidence-based decision support during supply disruptions
- **For fiscal policy**, it strengthens VAT integrity and reduces the VAT gap
- **For sustainability**, it enables precise measurement of embedded carbon flows and targeted decarbonization strategies that balance environmental and employment objectives
- **For corporate governance**, it offers, as secondary benefit, efficient compliance with due diligence directives without excessive administrative burdens.

The conclusion is therefore clear: Europe's competitiveness, sustainability, and security depend predominantly on mastering its own economic interconnections. Building an EU-wide, trusted supply-chain intelligence capability is not optional—it is an essential public infrastructure for the data age. It will allow Europe to move from reactive crisis management to proactive strategic foresight. The challenge is immense, but so is the reward: a resilient, sustainable and competitive European economy that can navigate global turbulence with confidence, grounded in evidence, guided by insight and strengthened by cooperation.

We close with the statement that the establishment of an effective *European Supply-Chain Intelligence Facility* will need strong and determined leadership and determination on the European level. Strategic institutions, such as the one envisioned, do not emerge bottom up. Missing but necessary EU legislation, the necessity to access critical data from individual member states and the size of the technical challenges make it impossible for individual member states to accomplish the task.

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ANNEX 1. DATA SOURCES DRIVING THE SUPPLY NETWORK REVOLUTION

Commercial datasets

Commercial business intelligence platforms such as *S&P Capital IQ*, *FactSet*, *CompuStat*, and *Bloomberg* provide structured information on firm-to-firm supply relationships, often compiled from SEC filings, press releases or voluntary reports. The largest among them, *S&P Capital IQ*, currently includes approximately 1.2 million firms and 1.4 million reported links [Chakraborty2020]. Smaller datasets such as *CompuStat* contain roughly 915 firms and 1033 links, while *FactSet* covers 6,714 firms and 40,861 supply relations [Mungo2022]. However, these commercial datasets suffer from severe limitations. Coverage is incomplete, with significant biases toward large listed firms in high-income economies—the USA in particular—and many links are outdated or unverified. Time validity is unclear: Are these relationships still active? Are they repetitive? For how long? Relationships are often not *weighted*—that is, they do not indicate how much economic value or volume is flowing between firms. Further, large-scale extraction is restricted due to licensing constraints. A well-cited case study illustrates this limitation vividly: mapping the supply chain of Honda required over 300 person-hours to identify just 10,000 companies and 50,000 relationships manually [Shao2018].

Credit rating and industry reporting systems

Some countries maintain firm-disclosed supplier datasets, such as Japan's *Tokyo Shoko Research Database*[14], where companies voluntarily report critical suppliers for credit rating purposes. This source alone covers around 1 million firms and 4 million links, making Japan one of the most transparent economies in terms of supply chain visibility.

Administrative datasets

A more comprehensive alternative is emerging from regulatory datasets, particularly value added tax (VAT) records, held by national administrations for fiscal monitoring. Because VAT systems record actual monetary transactions between firms, they allow precise reconstruction of domestic supply networks *at full population scale*. Countries such as Hungary, Spain, Belgium, Ecuador, Chile and Estonia already maintain VAT-based datasets that have been used to map entire national production systems with near-universal coverage, albeit only for *domestic* links; see [Pichler2023] for a recent overview. The need for harmonized EU-wide coverage is emphasized in the policy recommendations in section 5. Making use of such data for supply network reconstruction can lead to data protection issues and must be done in ways fully compliant with GDPR, see section 4.1.4.

Similarly, payment systems operated by central banks offer transaction-level visibility. Brazil, for instance, records real-time gross settlement data on inter-firm payments [Silva2021], while Japan and the Netherlands have used bank-level data to estimate supply flows; see [Fujiwara2021] and

14 Tokyo Shoko Research Database, https://www.tsr-net.co.jp/global/

[Ialongo2022], respectively. However, the challenge here lies in distinguishing economic supply relationships from financial transfers, making the estimation of supply networks non-trivial.

ANNEX 2. POSITIVE SIDE EFFECTS OF SUPPLY CHAIN INTELLIGENCE – POLICY TESTING IN SILICO

Since real-world supply networks can now be mapped and simulated in detail, policy does no longer have to rely on intuition, it can be *tested ex-ante*. [Stangl2024] provides one of the first demonstrations of this approach by using Hungarian VAT-based supply chain data to assess how different specific decarbonization strategies would affect employment and output.

The used framework first quantified each firm's systemic employment relevance—that is, how many jobs depend on its continued operation, directly or indirectly (not only in the affected firm itself but also along its supply chain) through supply relationships. This was then combined with firm-level CO_2 emission data, identifying companies that act as decarbonization leverage points. These are firms with *high emissions* but *low systemic importance in the economy*, where climate action can be implemented with minimal economic disruption and social cost.

The study analysed alternative decarbonization policy scenarios for reaching fixed emission reduction targets within relatively short time. The results showed that *identical emission reduction targets* could lead to dramatically different economic outcomes, depending on the strategic focus:

- **Prioritizing emissions only** (ignoring systemic relevance) led to **28% job loss** and **33% drop in economic output,** a politically impossible outcome.
- **Protecting jobs at each firm individually**, without considering their network impact, led to similarly **unacceptable outcomes**.
- **Prioritizing systemic economic importance** improved stability but still required removing many essential firms to meet emission targets.
- Only a **combined strategy balancing systemic relevance with emissions intensity reduced both emissions and economic losses**, limiting the employment and output decline to just **2%**.

The core insight here is that the same emission target can be socially "expensive" or "cheap" depending on where pressure (intervention) is applied in the network. Some of the largest emitters are *also* systemically essential, but *not all*. Given that substantial emission reduction targets should be reached in production networks over short time scales, policy should consider to not treat major emitters uniformly. The authors propose exploring supply-chain-sensitive CO_2 taxes, where taxation reflects *both* emission levels *and* systemic relevance. Such approaches could accelerate decarbonization while maintaining socio-economic stability. The calculation of such supply-chain-sensitive carbon taxes could be based on actual carbon flows derived from supply networks[15] which would eliminate strategic behaviour of firms to avoid it

15 Supply network information at the firm level allows one to monitor the carbon footprint of all individual firms within an economy, see [Stangl2025].

Technology now allows for complete, firm-level mapping of supply chains. The European Union can use this to build a Digital Twin of its economy, enabling it to simulate shocks, identify vulnerabilities, and enhance resilience. Developing this capability is essential for the EU's strategic autonomy as global competitors advance similar tools. We recommend establishing a dedicated legal and institutional framework to secure the Union's economic preparedness and competitive advantage.
This document was provided by the Economic Governance and EMU Scrutiny Unit at the request of the ECON Committee.

PE 779.855

ECTI/A/ECON-ED/IC/2025-017
Print ISBN 978-92-848-3140-1| doi: 10.2861/8451440| QA-01-25-249-EN-C
PDF ISBN 978-92-848-3139-5| doi: 10.2861/2906543| QA-01-25-249-EN-N